Jérôme Lin

Software Engineering Student — Seeking a 4-to-6-Month DevOps / Systems Internship

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TECHNICAL SKILLS

Systems & OS: Linux/UNIX internals, POSIX API, x86-64 Assembly
Languages: C, C++, POSIX Shell (Bash), Python, Java, SQL, C#, JavaScript/TypeScript, HTML/CSS
DevOps & Tools: Criterion (C Unit Testing), Shell Test Suites, Make / Makefiles, GCC/Clang, GDB, Valgrind **Networking & Databases:** TCP/IP, Sockets, DNS, HTTP/HTTPS, SQL (PostgreSQL, SQLite)

KEY TECHNICAL PROJECTS

POSIX Shell Interpreter (42sh) | *C, POSIX API, Linux, AST, Git* **EPITA SAÉ**

- Architected a compliant POSIX shell from scratch in C under strict coding norms (-Wall -Wextra -Werror).
- Handled process lifecycle, I/O redirection pipelines, and asynchronous signals (`SIGINT`, `SIGTERM`).
- Built a lexical analyzer and recursive descent parser producing an Abstract Syntax Tree (AST).

Minimake — Simplified Build System | *C99, Makefile Parsing, Process Scheduling* **EPITA SAÉ**

- Built a UNIX `make(1)` clone executing dependency graphs, pattern rules (%), and variable expansions.
- Parsed timestamps via `stat(2)` to trigger builds and run parallel child subshell executions.

Dynamic Memory Allocator (libmalloc) | *C, Linux Kernel API, mmap, Pthreads* **EPITA SAÉ**

- Implemented custom `malloc`, `free`, `calloc`, and `realloc` using `mmap(2)` and chunk metadata.
- Ensured thread safety via POSIX mutexes and injected into binaries using `LD_PRELOAD`.

OCR & Word Search Solver | *C, Image Processing, Neural Networks* **EPITA Prep Cycle**

- Implemented an end-to-end character recognition pipeline in C without external ML frameworks.
- Built image filtering algorithms (binarization, grid detection) and a custom multilayer perceptron for letter classification.

PROFESSIONAL EXPERIENCE

Saipem SA — *Engineering Intern (Drilling Digital Twin)* **June 2025 – July 2025**

- Developed a digital twin application representing an industrial drilling rig for real-time visualization.
- Hands-on experience in software lifecycles, cross-functional teamwork, and industrial safety.

Tommy Hilfiger — *Retail Operations & Sales Associate* **December 2025 – February 2026**

- Streamlined stock logistics, restocking, and visual merchandising adhering to brand standards.
- Handled international customer interactions, resolving inquiries and driving sales under peak store traffic.

EPITA — *School Ambassador* **2024 – Present**

- Represent the school at academic forums and open days; deliver presentations and advise prospective students on engineering programs.

EDUCATION

EPITA — School of Computer Science & Advanced Technologies Paris, France
Diplôme d'Ingénieur (Master of Science in Engineering, CTI accredited) **2024 – 2029**

- **International Preparatory Cycle** (English Section, 2024–2026).
- **Relevant Coursework:** Operating Systems (SYS), Networks (NET1), Relational Databases (SQL), C/Unix Programming (CNIX), Assembly (ASM), Software Architecture.

Lycée — *Baccalauréat Général (Mathematics & Physics-Chemistry)* **2024**
Highest Honors (Mention Très Bien)

LANGUAGES

French: Native | **English:** Fluent / C1 (TOEIC: 915/990, English-taught curriculum) | **German:** Intermediate (B1)